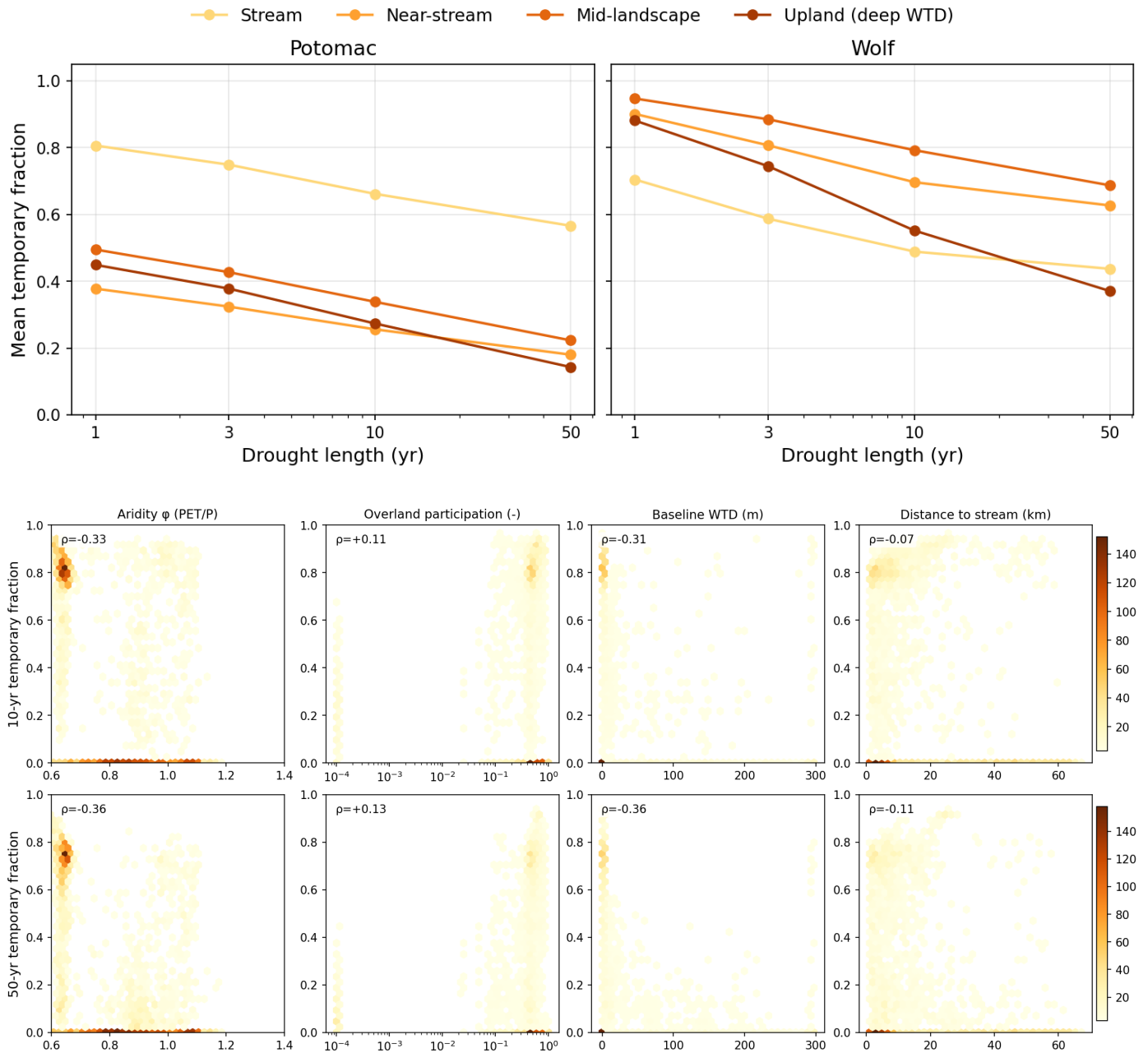


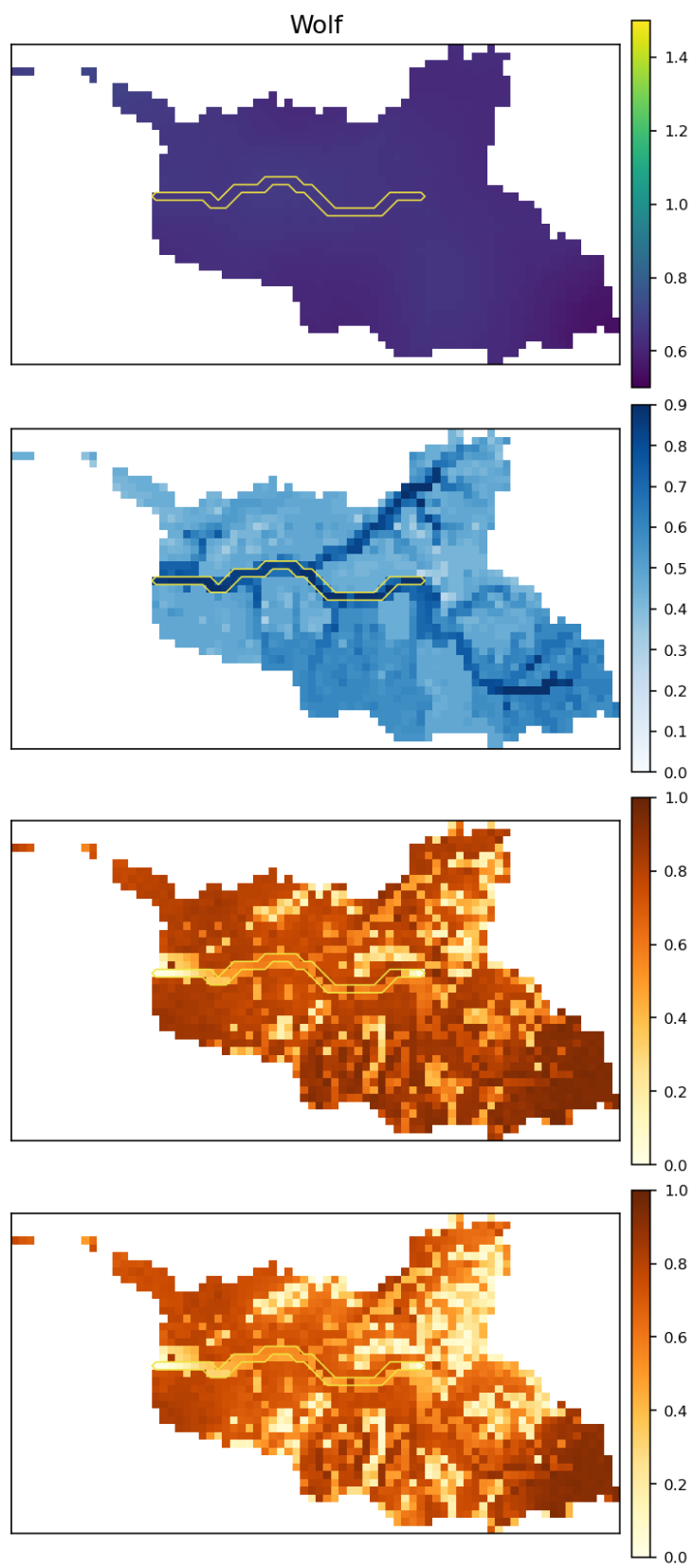
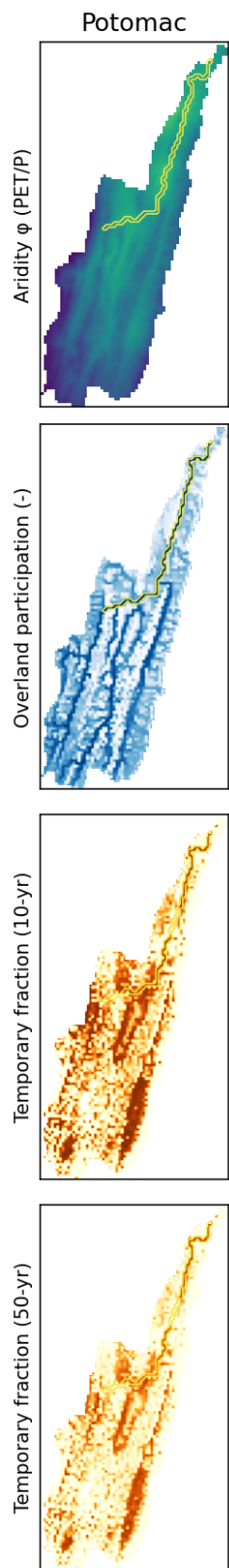
budyko temp persistent summary

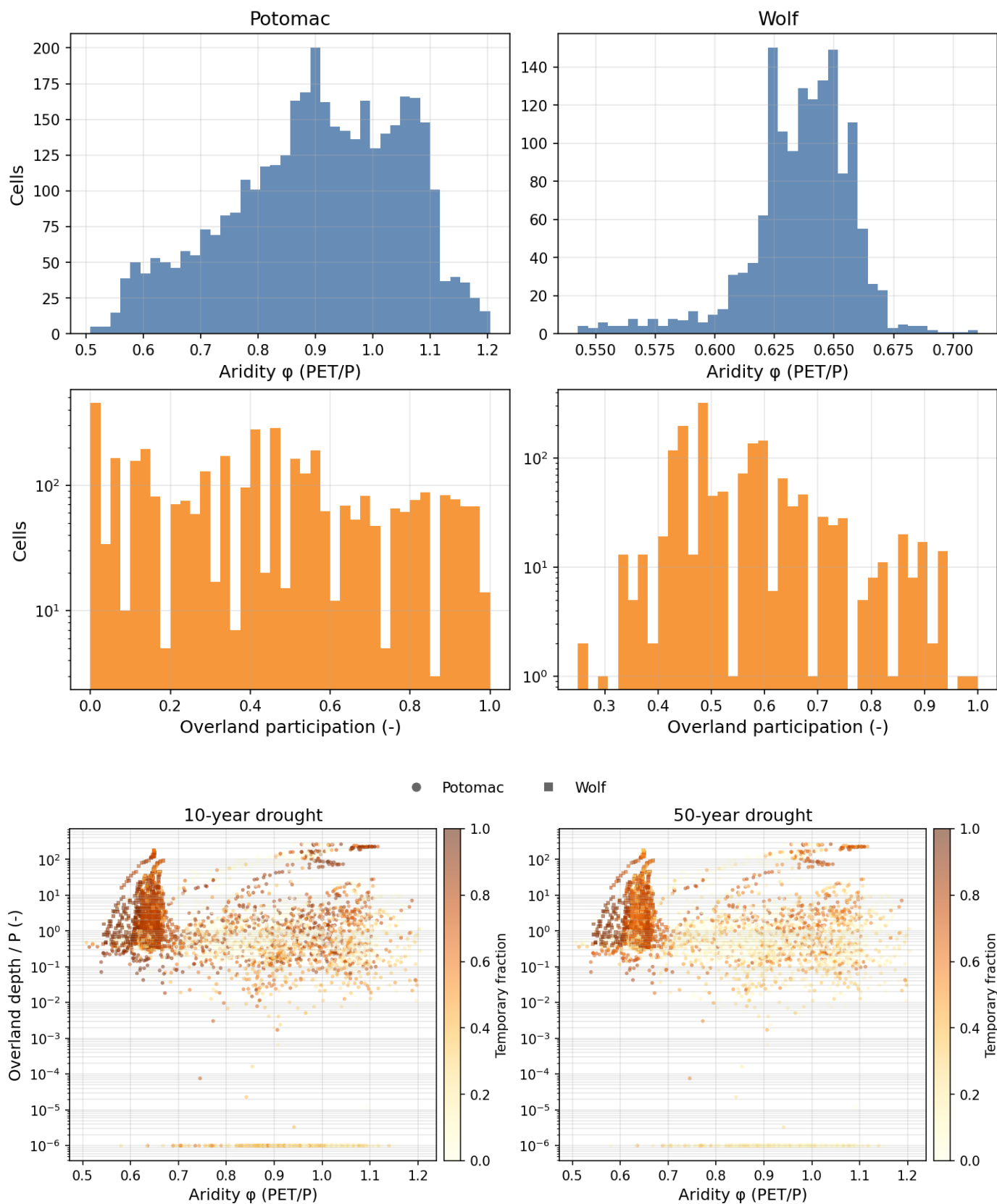
Local Budyko framing of temporary vs persistent deficits

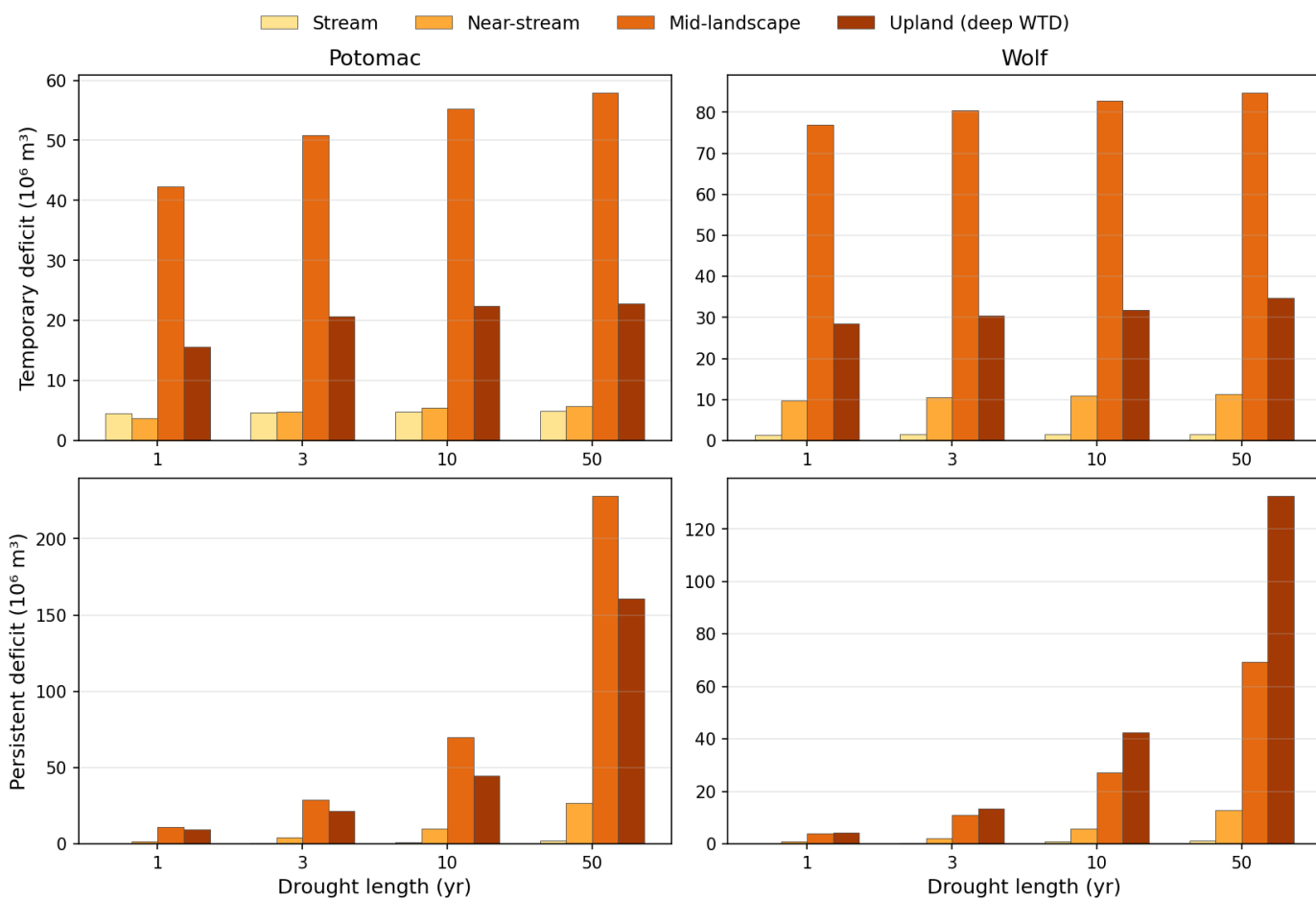
Script: budyko_temp_persistent.py

Figures: analysis/figures/budyko_*.png









Definitions

Quantity	Source
Temporary / persistent storage	Same as recovery analysis: recovered in recovery yr1 vs remaining
Temporary fraction f_{temp}	$S_{\mathrm{temp}}/(S_{\mathrm{temp}}+S_{\mathrm{pers}})$
Aridity ϕ	Hamon PET / CW3E APCP (average forcing year)
Overland participation	Fraction of 219 h steps with $Q_{\mathrm{ol}} > 1$ m ³ /h (late-spinup mean)
Strata	Stream (top 3% flow); near (≤ 2 km); upland (dist > 5 km + deep WTD tercile); mid = rest

Verdict

The temporary/persistent split behaves like a local, connectivity-weighted Budyko analogue, but not as a classical within-basin climate Budyko:

1. Between domains, climate aridity lines up with the story: Wolf is uniformly energy-limited ($\phi \approx 0.64$, high f_{temp}); Potomac sits near the energy/water threshold ($\phi \approx 0.90$, more persistent memory).
2. Within domains, ϕ barely varies (esp. Wolf), so cell-scale climate cannot explain temporary vs persistent. Baseline WTD + overland connectivity do - the "local energy-limited" end is shallow, flow-connected storage; the "local water-limited" end is deep-WTD upland storage.
3. Persistent deficit growth with drought length is strongest in uplands (x3.1-3.6 from 10->50 yr) and weakest in streams (x1.4-2.0) - the slow, recharge-limited branch.

Key numbers

Domain	ϕ [p05-p95]	P / PET (mm)	Stream f_{temp}	Upland f_{temp}	Persist growth 50/10 (upland)
Potomac	0.90 [0.62-1.11]	792 / 703	0.10 -> 0.57	0.19 -> 0.14	x3.60
Wolf	0.64 [0.60-0.66]	1532 / 974	0.49 -> 0.44	0.55 -> 0.37	x3.13

Within-domain Spearman vs f_{temp} (non-stream cells):

- Wolf: baseline WTD $\rho \approx -0.53$; overland flow/participation $\rho \approx +0.25-0.30$; ϕ weak.
- Potomac: all cellwise $|\rho| < 0.14$ for f_{temp} ; but persistent magnitude strongly anti-correlates with overland ($\rho \approx -0.48$).

Pooled hexbins mix domains, so ϕ - f_{temp} $\rho \approx -0.33$ is mostly the Wolf-vs-Potomac contrast.

Figures

File	What it shows
budyko_phi_vs_overland_hist.png	ϕ nearly uniform in Wolf; Potomac spans ~ 0.6 - 1.2 . Overland participation is the spatially rich axis.
budyko_maps_phi_overland_ftemp.png	Maps of ϕ , participation, f_{temp} (10 & 50 yr).
budyko_hexbin_ftemp_vs_predictors.png	f_{temp} vs ϕ , participation, WTD, stream distance.
budyko_space_ftemp.png	Local Budyko space ($\phi, Q_{\mathrm{ol}}/P$) colored by f_{temp} .
budyko_stratified_temp_persist_bars.png	Absolute temp/persist volume by stratum x drought length.
budyko_ftemp_by_stratum.png	Mean f_{temp} by stratum vs length (clearest landscape test).
budyko_spearman_ftemp.png	Within-domain Spearman heatmap.

Interpretation notes

- Absolute temporary volume is largest in mid-landscape because that is most of the area; stream cells are small but (on Potomac) have the highest **fraction** temporary.
- Wolf's temporary pulse is a wet-landscape feature (near/mid), not only the channel mask - consistent with a broader energy-limited, export-connected storage zone.
- Classical Budyko language fits best as: domain climate sets the baseline regime; local "Budyko" is WTD-overland position along the export vs recharge continuum.